INCH-POUND

MIL-DTL-28840/9E
20 April 2004
SUPERSEDING
MIL-C-28840/9D(EC)
25 June 1984

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 45°, CABLE SEALING AND SHIELD TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

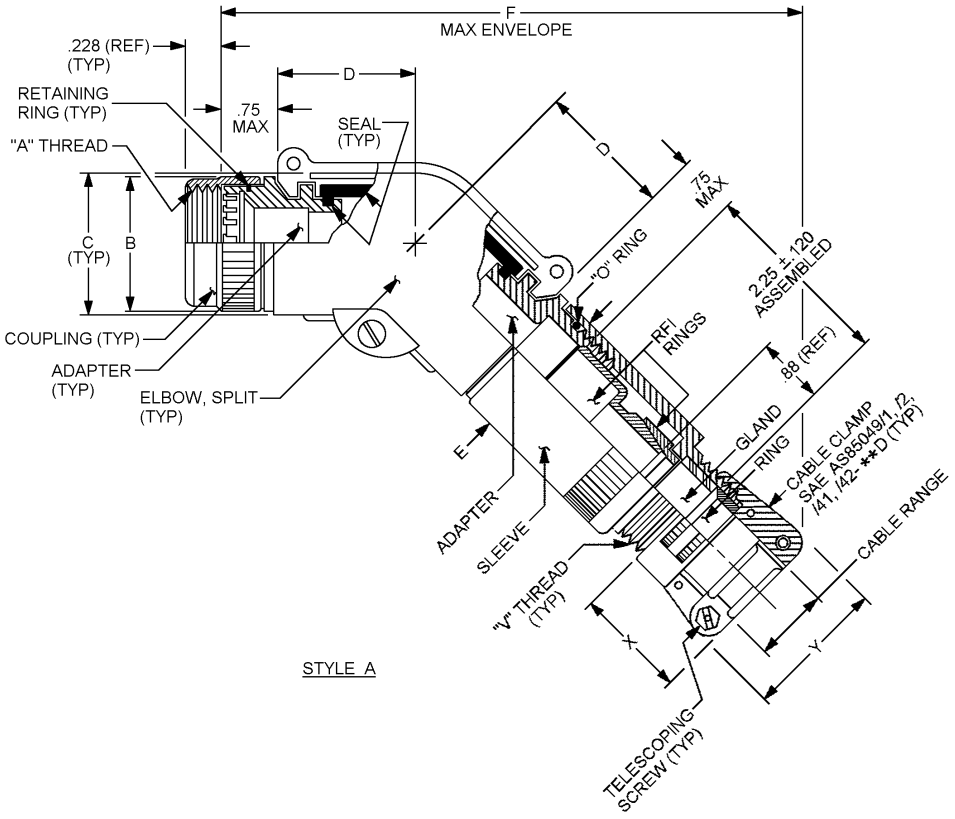
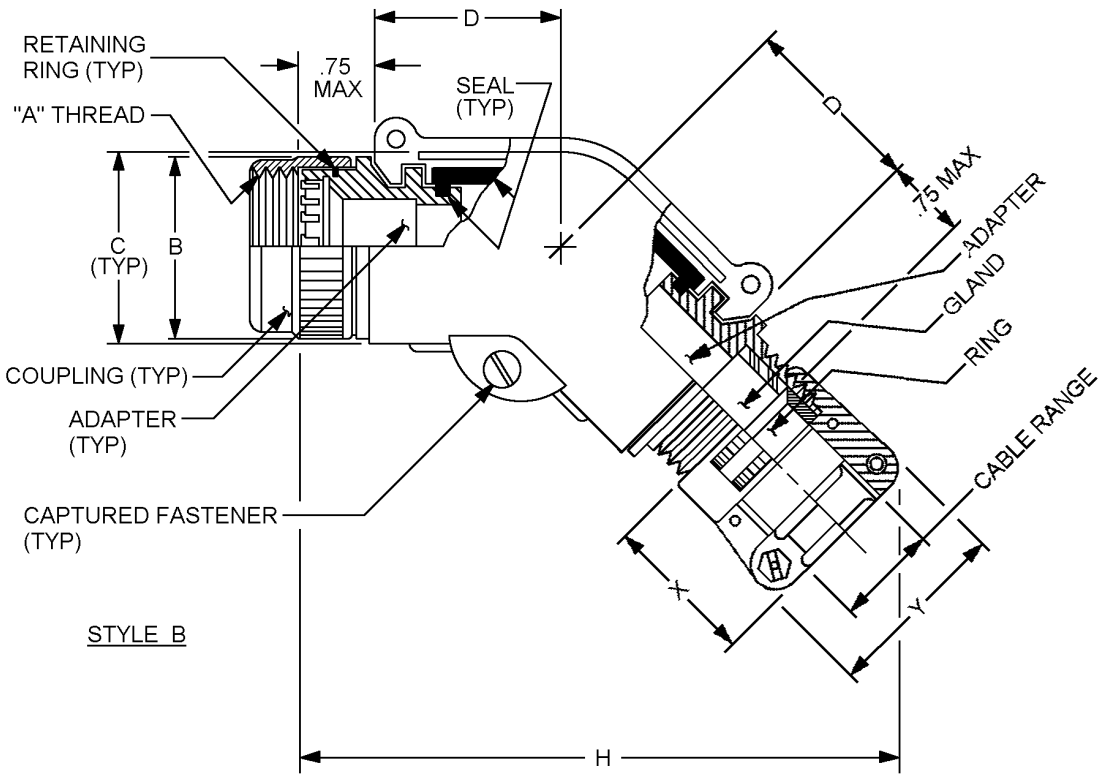


FIGURE 1. Dimensions and configurations.

AMSC N/A FSC 5935

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Inches	mm
.12	3.05
.228	5.79
.75	19.05
.88	22.35
2.25	57.15

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:35Z
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Back shell size (see note 5)	Dash no.	Designator (see note 3)	A thread	B max dia	C max dia	D max
A	01	A (11)	.750-20 UNEF	1.028 (26.11)	1.031 (26.19)	1.031 (26.19)
A	02	B (13)	.875-20 UNEF	1.141 (28.98)		
B	03	B (13)				
A	04	C (15)	1.00-20 UNEF	1.263 (32.08)	1.281 (32.54)	1.156 (29.36)
B	05	C (15)				
C	06	C (15)				
D	07	C (15)			1.513 (38.89)	1.281 (32.54)
A	08	D (17)	1.125-18 UNEF	1.387 (35.23)		
B	09	D (17)				
C	10	D (17)				
A	11	E (19)	1.250-18UNEF	1.513 (38.43)		
B	12	E (19)			1.781 (45.24)	1.406 (35.71)
A	13	F (23)	1.438-18 UNEF	1.703 (43.51)		
B	14	F (23)				
C	15	F (23)				
A	16	G (25)	1.562-18 UNEF	1.825 (46.35)		
B	17	G (25)				
C	18	G (25)			2.031 (51.59)	1.531 (38.89)
D	19	G (25)				
A	20	H (29)	1.875-16 UN	2.143 (54.43)		
B	21	H (29)				

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 5)	Dash no.	Designator (see note 3)	Cable range		E max	F max style A	H max style B
			max	min			
A	01	A (11)	.312 (7.92)	.188 (4.77)	1.156 (29.36)	5.879 (149.33)	4.199 (106.65)
A	02	B (13)	.312 (7.92)	.188 (4.77)		5.879 (149.33)	4.199 (106.65)
B	03	B (13)	.438 (11.12)	.281 (7.14)		5.932 (150.67)	4.252 (108.00)
A	04	C (15)	.312 (7.92)	.188 (4.77)	1.281 (32.54)	60.92 (154.74)	4.412 (112.06)
B	05	C (15)	.436 (11.12)	.281 (7.14)		6.146 (156.11)	4.466 (113.44)
C	06	C (15)	.562 (14.27)	.344 (8.74)		6.202 (157.53)	4.522 (114.86)
D	07	C (15)	.750 (19.05)	.438 (11.12)	1.406 (35.71)	6.533 (165.94)	4.853 (123.27)
A	08	D (17)	.625 (15.87)	.375 (9.62)		6.413 (162.89)	4.733 (120.22)
B	09	D (17)	.750 (19.05)	.438 (11.12)		6.533 (165.94)	4.853 (123.27)
C	10	D (17)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.636 (168.55)	4.956 (125.88)
A	11	E (19)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	6.413 (162.89)	4.733 (120.22)
B	12	E (19)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.166 (131.22)
A	13	F (23)	.750 (19.05)	.438 (11.12)	1.406 (25.71)	6.747 (171.37)	5.067 (128.70)
B	14	F (23)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.166 (131.22)
C	15	F (23)	1.388 (30.17)	.812 (20.62)	1.781 (45.24)	7.103 (180.42)	5.423 (138.23)
A	16	G (25)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.156(130.96)
B	17	G (25)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)	7.103 (180.42)	5.4239 (138.23)
C	18	G (25)	1.250 (31.75)	.875 (22.23)		7.316 (185.83)	5.636 (143.15)
D	19	G (25)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)	7.425 (188.60)	5.745 (145.92)
A	20	H (29)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)	7.316 (185.83)	5.636 (143.15)
B	21	H (29)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)	7.425 (188.60)	5.745 (145.92)

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:35Z
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Back shell size (see note 5)	Dash no.	Designator (see note 3)	V thread class	X ref dim	Y ref dim
A	01	A (11)	.625-24 UNEF	1.281 (32.54)	.938 (23.82)
A	02	B (13)	.625-24 UNEF		.938 (23.82)
B	03	B (13)	.750-20 UNEF		1.125 (28.57)
A	04	C (15)	.625.24 UNEF		.938 (23.82)
B	05	C (15)	.750-20 UNEF		1.125 (28.57)
C	06	C (15)	.875-20 UNEF		1.312 (33.32)
D	07	C (15)	1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
A	08	D (17)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
B	09	D (17)	1.188-18 UNEF	1.312 (33.32)	1.395 (40.62)
C	10	D (17)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
A	11	E (19)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
B	12	E (19)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
A	13	F (23)	1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
B	14	F (23)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
C	15	F (23)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
A	16	G (25)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
B	17	G (25)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
C	18	G (25)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
D	19	G (25)	2.000-16 UNS	1.625 (41.28)	2.343 (59.51)
A	20	H (29)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
B	21	H (29)	2.000-18 UNS	1.625 (41.28)	2.343 (59.51)

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 5)	Dash no.	Designator (see note 3)	A thread	B max dia	C max dia	D max
A	22	J (33)	2.062-16 UNEF	2.329 (59.16)	2.531 (64.29)	1.781 (45.24)
B	23	J (33)				
B	24	A (11)	.750-20 UNEF	1.028 (26.11)	1.031 (26.19)	1.031 (26.19)
C	25	A (11)			1.281 (32.54)	1.156 (29.36)
C	26	B (13)	.875-20 UNEF	1.141 (28.98)	1.281 (32.54)	1.156 (29.36)
D	27	B (13)			1.531 (38.89)	1.281 (32.54)
E	28	C (15)	1.000-20 UNEF	1.263 (32.08)		

Back shell size (see note 5)	Dash no.	Designator (see note 3)	Cable range		E max dim	F max style A	H max style B
			Max	Min			
A	22	J (33)	1.380 (35.05)	1.000 (25.40)	2.281 (57.94)	7.852 (199.44)	6.127 (156.77)
B	23	J (33)	1.625 (41.28)	1.250 (31.75)		6.454 (163.93)	6.454 (163.93)
B	24	A (11)	.438 (11.12)	.281 (7.14)	1.156 (29.36)	5.932 (150.67)	4.252 (108.00)
C	25	A (11)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	6.202 (157.53)	4.522 (114.86)
C	26	B (13)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	6.202 (157.53)	4.522 (114.86)
D	27	B (13)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	6.413 (162.89)	4.733 (120.22)
E	28	C (15)	.625 (15.87)	.375 (9.50)		6.413 (162.89)	4.733 (120.22)

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 5)	Dash no.	Designator (see note 3)	V thread class	X ref dim	Y ref dim
A	22	J (33)	2.000-18 UNS	1.625 (41.28)	2.343 (59.51)
B	23	J (33)	2.250-16 UNS	1.900 (48.26)	2.750 (69.85)
B	24	A (11)	.750-20 UNEF	1.281 (32.54)	1.125 (28.57)
C	25	A (11)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
C	26	B (13)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
D	27	B (13)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
E	28	C (15)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Shell sizes are provided within parentheses for information and are not a part of the designator.
4. Dimensions apply after plating.
5. Designator of backshell size within each shell size.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS

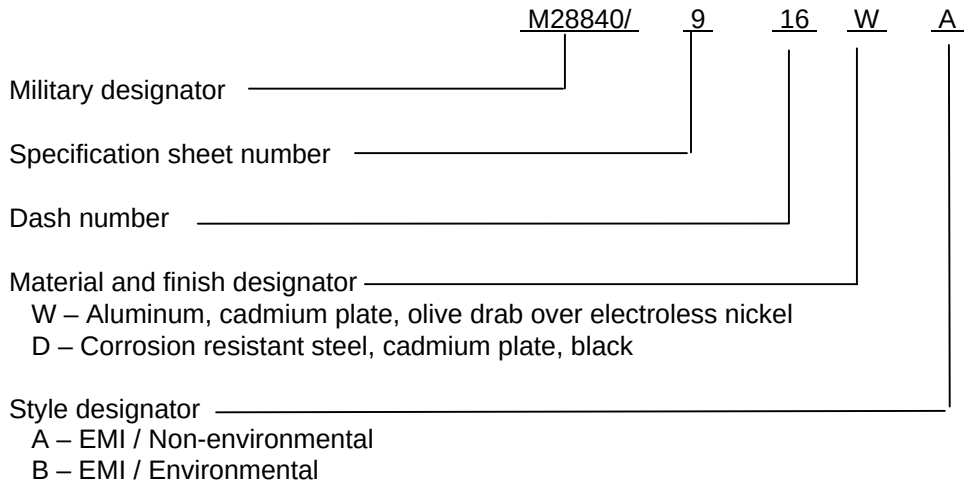
Dimensions and configurations: See figure 1.
Interface dimensions shall conform to MIL-DTL-28840.

Material: Shall be aluminum alloy in accordance with SAE-AMS-QQ-A-225, ASTM B85, ASTM B26/B26M, or ASTM B108. Corrosion resistant steel shall be in accordance with SAE-AMS-QQ-S-763, class 316.

Finish: Aluminum components shall be cadmium plate, olive drab over electroless nickel. Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivate in accordance with SAE-AMS-QQ-P-35.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:



Qualification: Two samples in each shell size for which qualification is desired and shall be tested in accordance with MIL-DTL-28840.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Reference documents. In addition to MIL-DTL-28840, this document references the following:

ASTM B26/B26M	SAE-AMS-QQ-S-763	SAE AS85049/1
ASTM B85	SAE-AMS-QQ-A-225	SAE AS85049/2
ASTM B108	SAE-AMS-QQ-P-35	SAE AS85049/41
		SAE AS85049/42

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CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4416-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online data at www.dodssp.daps.mil.

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